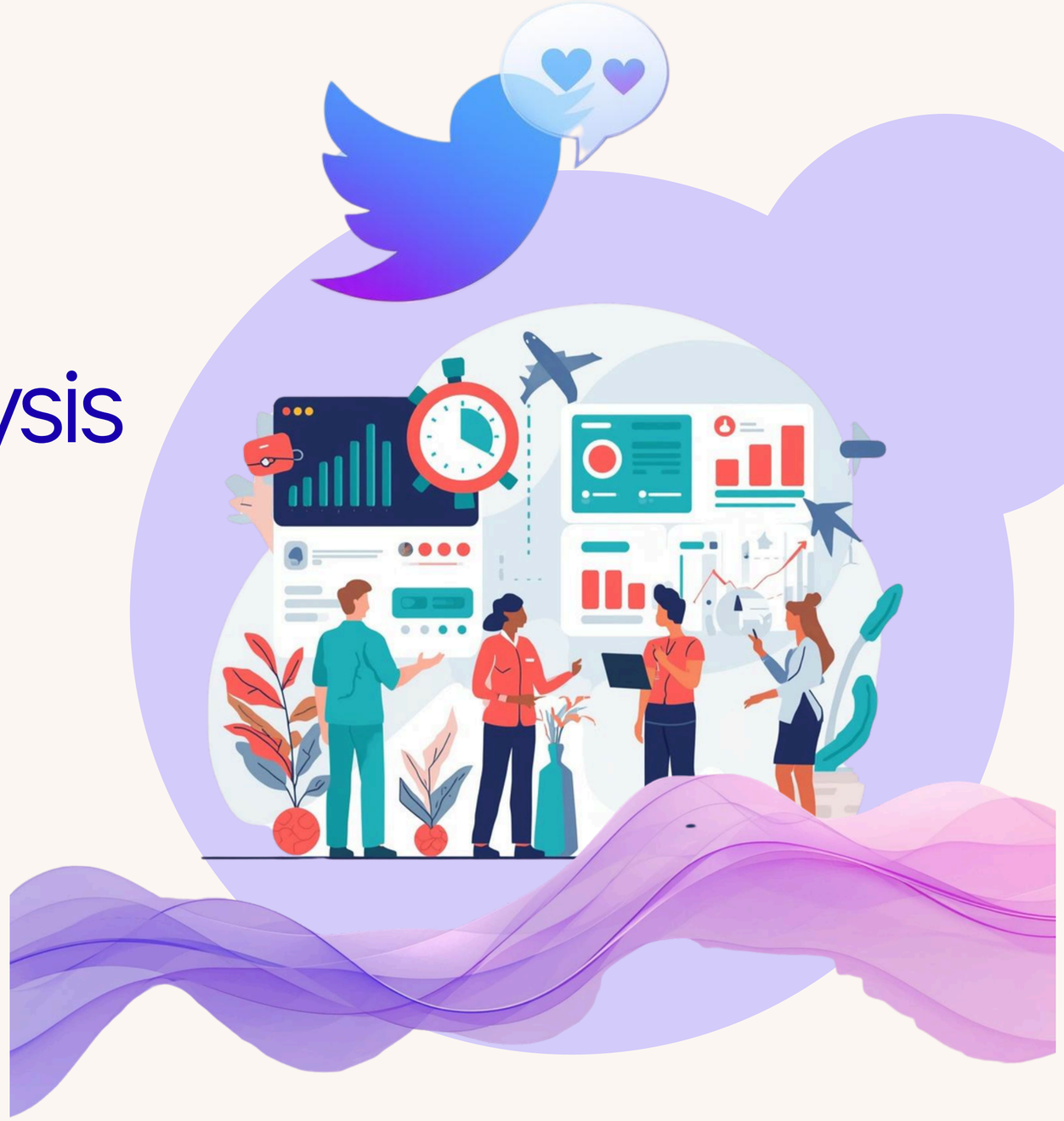
Airline Sentiment Analysis on Twitter



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Problem & Objective

The Problem

Airlines receive thousands of tweets every day, and there is no scalable solution for processing them. This creates a significant backlog, as manual reviews are slow and prone to bias, ultimately resulting in critical patterns being overlooked. Valuable customer feedback is wasted, which could otherwise inform service improvements and enhance customer satisfaction.

Our Objective

We propose an end-to-end NLP pipeline that classifies every tweet as Negative, Neutral, or Positive. By utilizing supervised Logistic Regression and unsupervised K-Means clustering, we can efficiently analyze sentiment in real-time. This approach allows airlines to harness customer feedback proactively, driving improvements in service and addressing passenger complaints effectively.



Dataset Overview

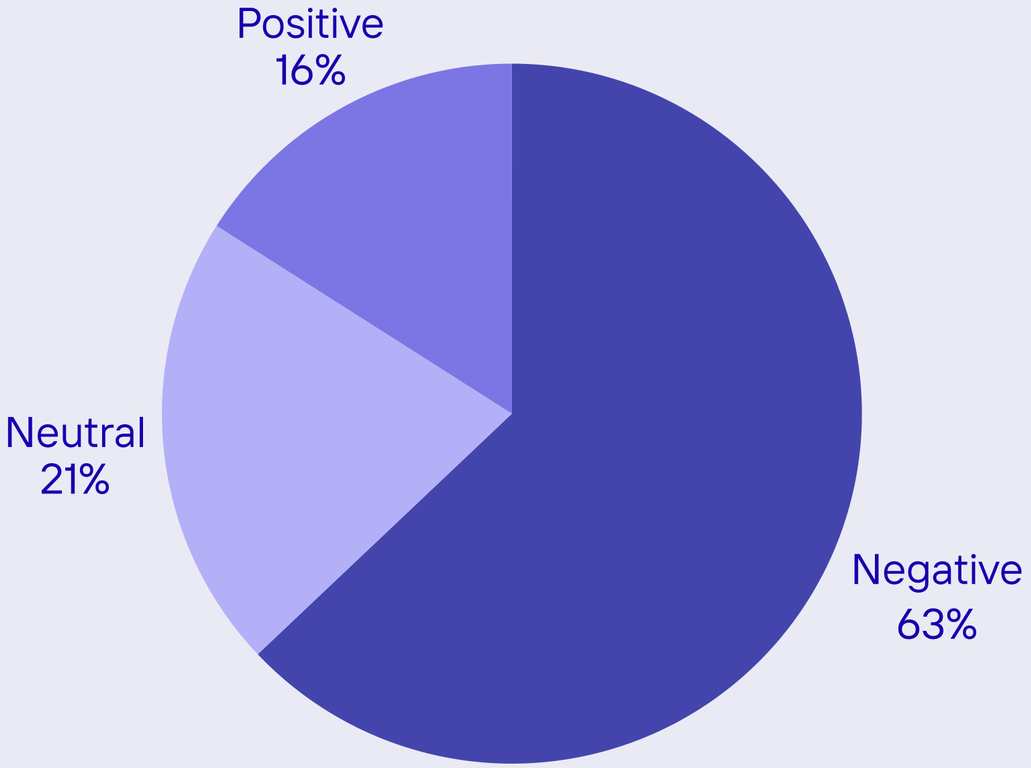
Tweet Volume : 14,640 tweets collected from Kaggle dataset.

Airlines Covered : 6 major U.S. airlines covered: United, American, Southwest, Delta, US Airways, and Virgin America.

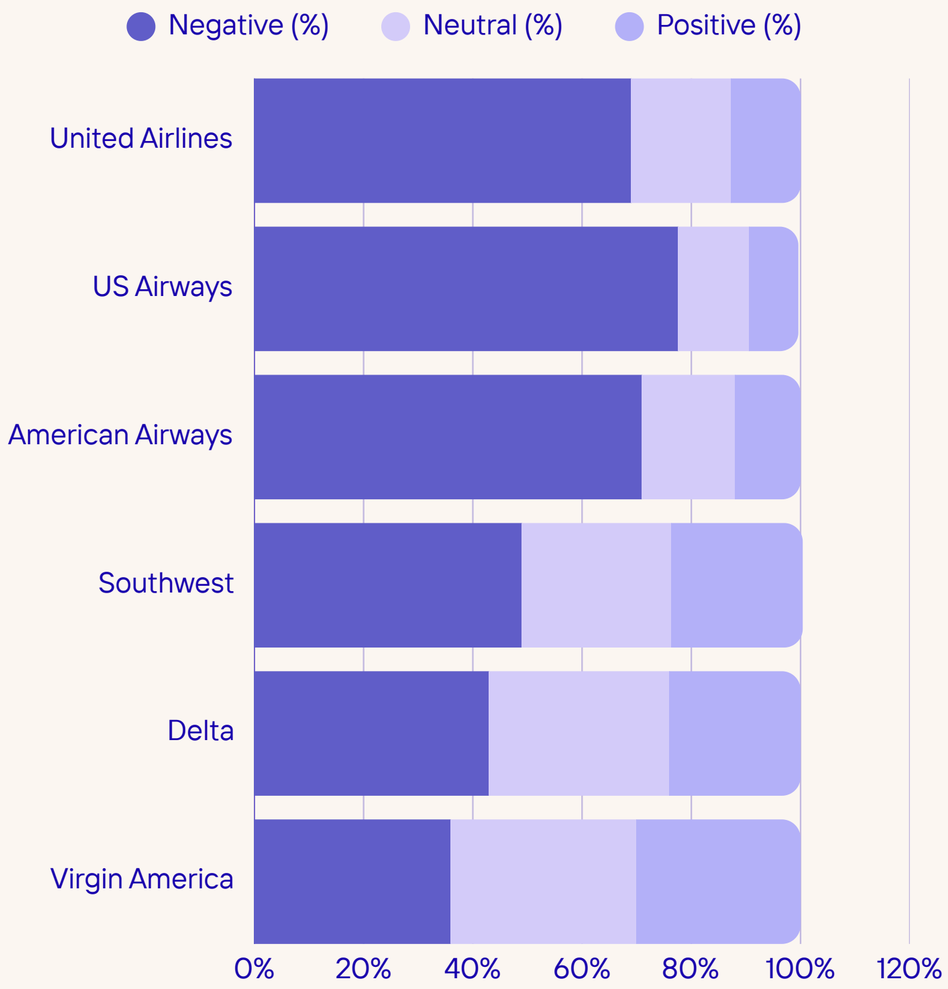
Sentiment Classes : 3 sentiment classes: Negative, Neutral, Positive.

Source : https://www.kaggle.com/datasets/zahranusratt/us-airline-twitter-sentiment-analysis-dataset

Sentiment Distribution



Key Insight:
Negative sentiment dominates airline Twitter discourse, reflecting vocal dissatisfaction among passengers. This imbalance highlights that customers are more likely to voice complaints than praise on social media platforms, making sentiment analysis crucial for identifying service improvement opportunities.

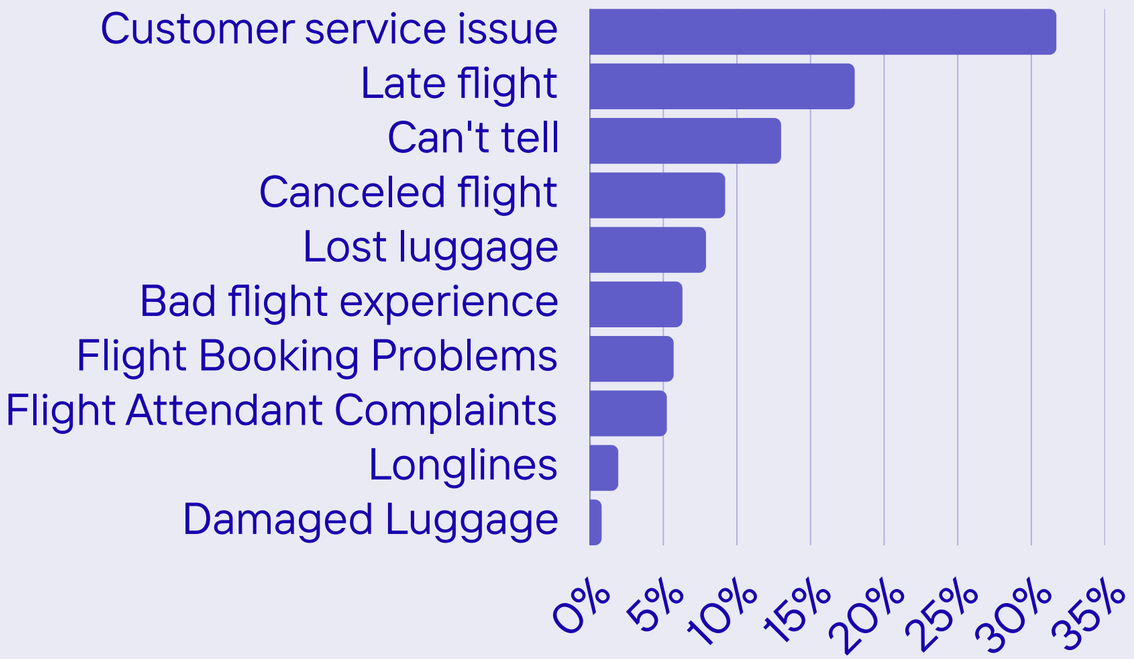


Sentiment by Airline Comparison

Customer satisfaction varies across airlines:

→ United Airlines and US Airways show the highest concentration of negative sentiment, suggesting they face more service-related issues.

→ Southwest and Delta have moderate negativity, while Virgin America stands out with the most balanced sentiment, suggesting better customer satisfaction



Top Reasons for Negative Sentiment

Negative sentiment is mainly driven by customer service issues (31.7%), followed by flight delays and cancellations. These results show that improving communication and service quality is just as important as operational performance for reducing customer dissatisfaction.

Text Preprocessing

```python
def preprocess(text):
    text = re.sub(r'@\w+', '', text)
    text = re.sub(r'http\S+', '', text)
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    text = text.lower()
    tokens = text.split()
    tokens = [t for t in tokens if t not in STOPWORDS and len(t) > 2]
    return tokens

df['tokens'] = df['text'].apply(preprocess)
```

Step 1: Remove @mentions and URLs

Clean raw tweet text by stripping user handles and web links that add noise to analysis.

Step 2: Convert to Lowercase

Standardize all text to lowercase to ensure consistent word matching and reduce vocabulary size.

Step 3: Tokenization

Split text into individual words (tokens) to enable word-level analysis and counting.

Step 4: Remove Stopwords

Filter out common words like "the", "is", "and" to focus on meaningful sentiment-bearing terms.

TF-IDF Configuration

TF-IDF Configuration

TF-IDF was used to convert tweets into numerical features by weighting words based on their importance in each tweet versus the whole dataset. it reduced noise by down-weighting repeated words, removing very rare words, and excluding overly common words. This produced a clean vocabulary of about 3,700 meaningful terms used as model inputs

Word Clouds by Sentiment

Negative Sentiment

Dominant words reveal passenger frustrations:

- "delayed" - flight timing issues
- "cancelled" - disrupted travel plans
- "lost" - baggage problems
- "waiting" - poor service response

These terms appear frequently in complaint tweets.



Word Clouds by Sentiment

Neutral Sentiment

General airline-related vocabulary without strong emotion:

- "flights" - basic travel reference
- "thank" - expressions of appreciation
- "need" -general requests or inquiries



Word Clouds by Sentiment

Positive Sentiment

Appreciative language showing satisfaction:
- "great" - quality praise
- "amazing" - exceptional experiences
- "thank" - gratitude expressions
- "love" - strong positive emotion

Reflects memorable positive experiences.



Methods Overview

SUPERVISED LEARNING

Model: Logistic Regression

Features: TF-IDF (3,700 terms)

Class imbalance handled via balanced class weighting

Evaluated with: Accuracy, Balanced Accuracy, Macro F1, Per-class F1

Requires labeled data: YES

UNSUPERVISED LEARNING

Model: K-Means Clustering

Dim. Reduction: PCA → 50 components

Optimal K selected via Elbow Method (WCSS)

Evaluated with: Silhouette Score, Cluster Purity

Requires labeled data: NO

Model: Logistic Regression

→ Logistic Regression was applied on the labeled tweet dataset using TF-IDF features.

→ In this setup, each tweet was transformed into a numerical representation based on the importance of its words within the dataset.

→ The model was trained directly on this representation, using the original class distribution without any adjustment.

```
lr_before = LogisticRegression(
    C=1.0, max_iter=1000, solver='lbfgs',
    multi_class='multinomial', random_state=42
    # No class_weight → model treats all classes equally
    # → heavily biased toward Negative (63% of data)
)
lr_before.fit(X_train_tfidf, y_train)
y_pred_before = lr_before.predict(X_test_tfidf)
m_before = get_metrics(y_test, y_pred_before)

print(f"\n  Accuracy          : {m_before['Accuracy']:.4f}")
print(f"  Balanced Accuracy : {m_before['Bal. Accuracy']:.4f}")
print(classification_report(y_test, y_pred_before, target_names=class_names))
```

Class Imbalance

The Skew

The dataset shows a strong sentiment imbalance, with 62.7% of tweets labeled as Negative.

This overrepresentation of complaints introduces a majority class bias, which reduced the model performance and weakened the learning of minority classes.



The Fix

The Approach

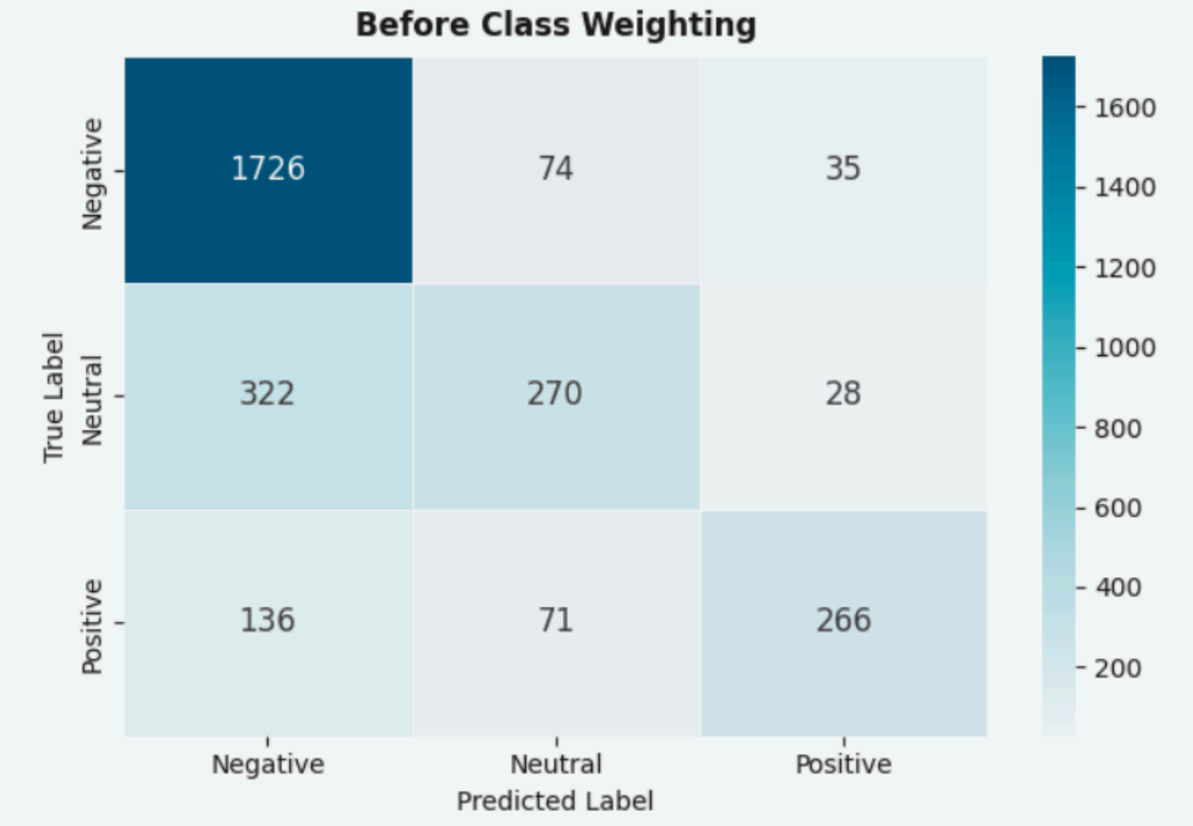
To correct the imbalance in our dataset, we apply class_weight='balanced' during model training.
This technique ensures each class receives a weight inversely proportional to its frequency in the dataset, allowing the model to focus more on minority classes while still considering majority classes effectively for better prediction accuracy.

The Weights

- The weights assigned to each class in this model are as follows:
- Negative class is downweighted to 0.529 to reduce dominance in training
- Neutral class is upweighted to 1.574 to improve learning of minority patterns
- Positive class is upweighted to 2.067 to further strengthen its representation

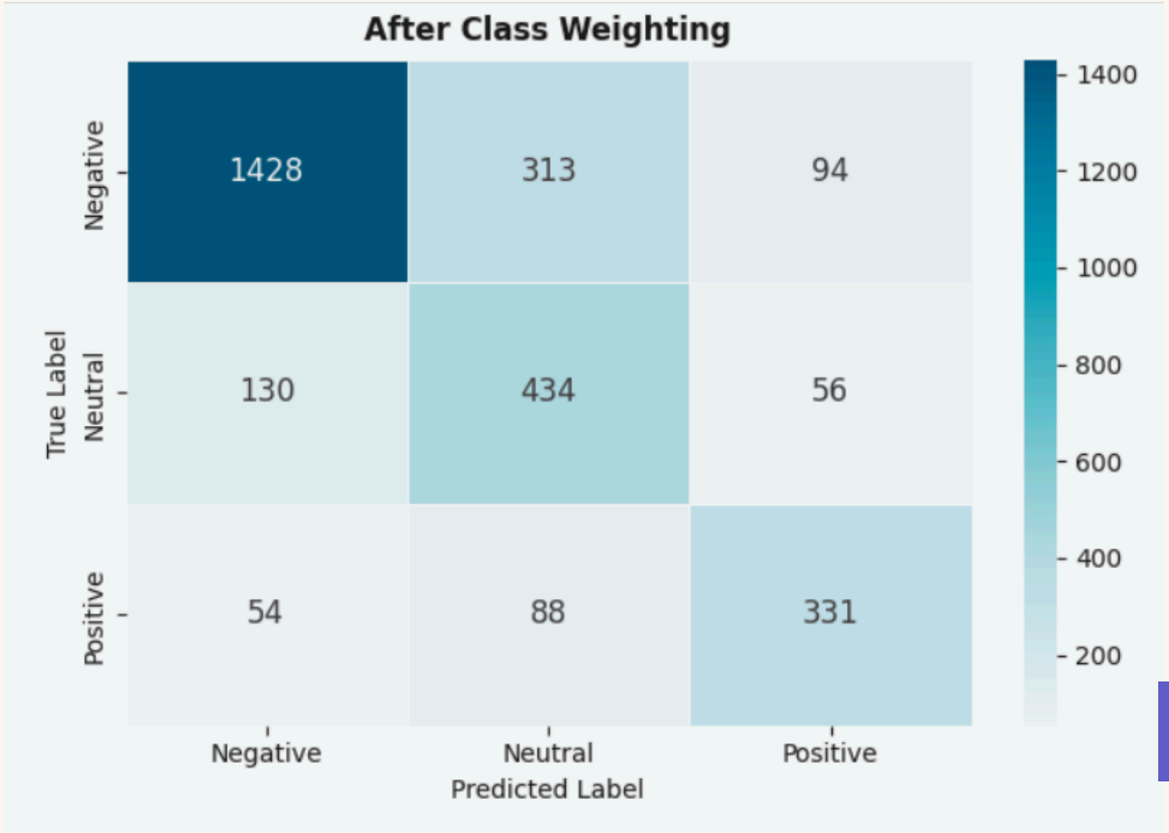
This balance helps the model learn all three sentiments fairly, addressing the class distribution issue.

confusion matrices



- Strong bias toward the majority class (Negative)
- Frequent misclassification of Neutral tweets as Negative (322 cases)
- Significant misclassification of Positive tweets as Negative (136 cases)
- Poor discrimination between sentiment classes

- Neutral misclassification reduced from 322 to 130
- Positive misclassification reduced from 136 to 54
- Better recognition of all three sentiment classes
- Reduced bias toward the Negative class
- Overall increase in sentiment classification accuracy



Results & Performance

Before weighting

Biased

Accuracy	77.25% (misleading)
Balanced accuracy	64.62%
Macro F1	68.13%
Neutral F1	52.17%
Positive F1	66.33%

After weighting

Balanced

Accuracy	74.90% (misleading)
Balanced accuracy	72.60% +8pp
Macro F1	70.63%
Neutral F1	59.66% +7.5pp
Positive F1	69.39% +3.1pp

On imbalanced data: balanced accuracy and macro F1 are the only metrics that tell the truth.

PCA + K-means

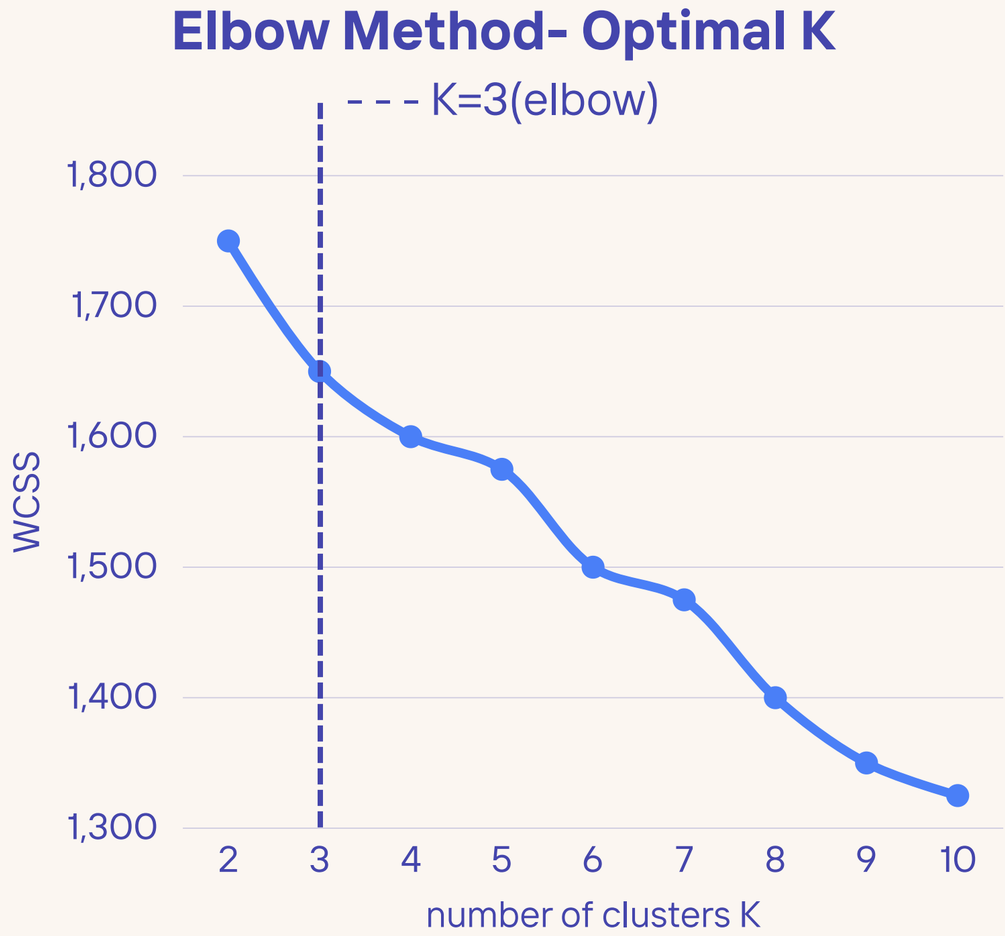
Dimensionality Reduction via PCA

→Principal Component Analysis (PCA) was used to reduce the high-dimensional TF-IDF space into a smaller set of 50 principal components, preserving the most important information while removing noise.

→This makes the data more suitable for efficient clustering and helps K-Means perform better in a lower-dimensional space.

Finding K:

→ To select the optimal number of clusters, the Elbow Method was used by plotting WCSS against different values of K. The elbow point at K = 3 indicates the best balance, matching the three sentiment classes and improving clustering interpretability.

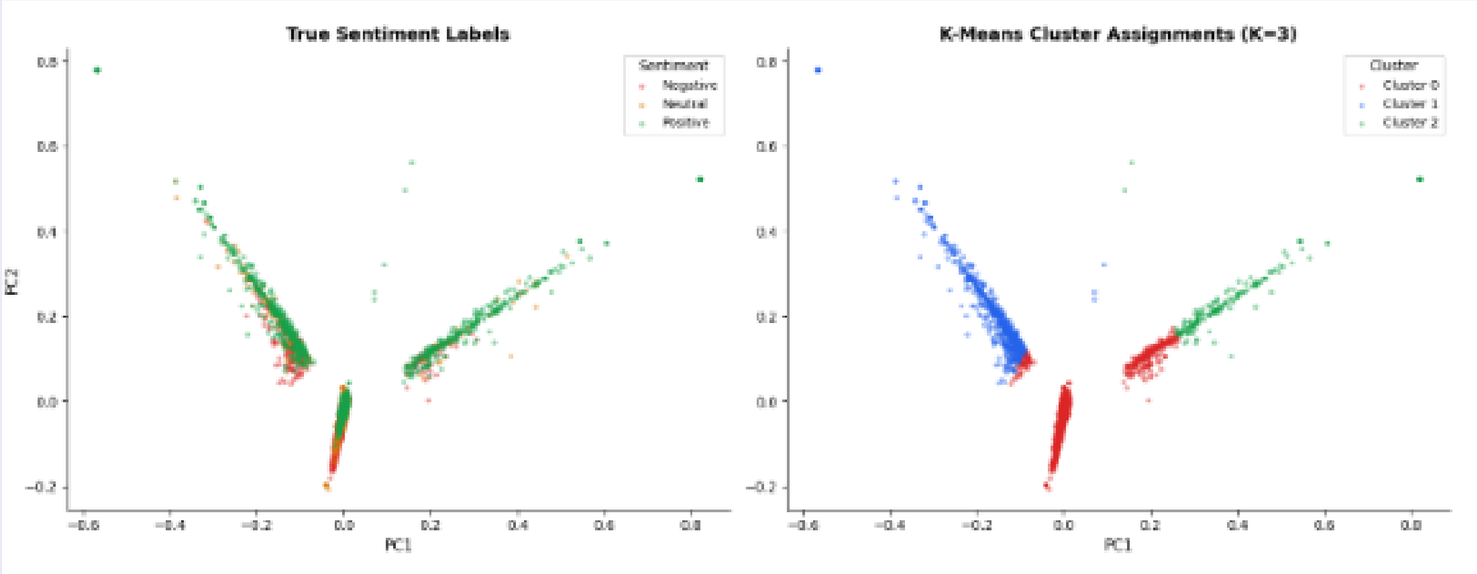


K-Means Results

Evaluation Metrics:

K-Means clustering achieved a Cluster Purity of 66.5%, indicating that most tweets were grouped into the correct sentiment clusters.

A Silhouette Score of 0.218 suggests a moderate but meaningful cluster structure, enabling insights without labeled data.



Cluster Profiles:

Cluster 0: Mostly Negative sentiment, with keywords like “cancelled” and “delayed”.

Cluster 1: Mixed Negative and Neutral sentiment, containing more ambiguous language.

Cluster 2: Predominantly Positive sentiment, featuring words like “thank”, “great”, and “love”.

Supervised VS Unsupervised

Logistic regression

Supervised

Accuracy	74.9%
Balanced accuracy	72.6%
Macro F1	70.6%

- Requires labeled data
- Highly interpretable
- Production-ready

Best for classifying new tweets at scale.

K-Means clustering

Unsupervised

Cluster purity	66.5%
Silhouette score	0.218
—	

- No labels needed
- Discovers hidden structure
- Exploratory by nature

Best for understanding unknown patterns.

One classifies. The other explores. Together: a complete picture of airline sentiment.

Business Insights & Conclusion

The real cost of doing nothing

62% of all passenger tweets are negative — and most complaints are entirely preventable.

Customer service issues	31.7%
Late flights	18.1%
Cancelled flights	9.2%

Every unresolved tweet is a public reputation hit.

The opportunity

Virgin America proves it is possible — the highest positive sentiment of all 6 airlines.

- Most negative: **United & US Airways**
- Most positive: **Virgin America**

The gap is not luck — it is operational choices.

Fix customer service + punctuality → fix the majority of your social media reputation.

Thank You!